THE HOMOLOGY OF THE EILENBERG-MACLANE OBJECT IS CONCENTRATED IN ONE DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-homology-eilenberg-maclane (source lines 3933-3940); the proof environment begins two lines below it and is its own proof, closing at source line 3969.

Editorial note. The proof invokes two author results of the same chapter by label (the extension construction and the acyclicity of the extension complex); both are retained as context spans and the generator expands the context transitively. The final sentence names a Homology-chapter result (homology-lemma-long-exact-sequence-chain) whose target carries a canonical project tag, so it is auto-declared as a non-load-bearing source pointer and resolves inside the assembled package.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Objects chapter, counted once: the homology of the Eilenberg-MacLane object K(A,k) is A in degree k and vanishes elsewhere.

Allowed standard prerequisites (editorial).

- The Eilenberg-MacLane object K(A,k). The simplicial object concentrated in degree k with value A, built from the canonical extension construction (the author's lemma lemma-extension, retained as context).. It is the object whose associated complex is studied (source lines 3935-3939).
- Acyclicity of the extension object's complex. The complex attached to the extension object E is acyclic (the author's lemma lemma-homology-extension, retained as context).. It is applied in the induction step to kill the third term (source lines 3960-3966).
- Long exact sequence of homology. The long exact homology sequence attached to a short exact sequence of complexes.. It transfers the computation across the short exact sequence relating consecutive Eilenberg-MacLane objects (source lines 3966-3968).

Source reference. Original lemma 0111: sources/stacks-chow/tags.txt, source lines 1347–1347. Cross-chapter pointer whose target carries the canonical tag 0111 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

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Source reference. Original lemma 0110: sources/stacks-chow/tags.txt, source lines 1346–1346. Cross-chapter pointer whose target carries the canonical tag 0110 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let A be an object of $\mathcal{A}$ and let k be an integer ≥ 0. Consider the simplicial object E defined by the following rules*

1. *$E_n = \bigoplus_\alpha A$, where the sum is over $\alpha : [n] \to [k+1]$ whose image is either $[k]$ or $[k+1]$.*
2. *Given $\varphi : [m] \to [n]$ the map $E_n \to E_m$ maps the summand corresponding to α via id_A to the summand corresponding to $\alpha \circ \varphi$, provided $\mathrm{Im}(\alpha \circ \varphi)$ is equal to $[k]$ or $[k+1]$.*

Then there exists a short exact sequence

$$0 \to K(A,k) \to E \to K(A,k+1) \to 0$$

which is term by term split exact.

Proof. The maps $K(A,k)_n \to E_n$ resp. $E_n \to K(A,k+1)_n$ are given by the inclusion of direct sums, resp. projection of direct sums which is obvious from the inclusions of index sets. It is clear that these are maps of simplicial objects. □

Lemma 0.2. *Let $\mathcal{A}$ be an abelian category. Let A be an object of $\mathcal{A}$ and let k be an integer. Let E be the object described in Lemma 0.1. Then the complex $s(E)$ is acyclic.*

Proof. For a morphism $\alpha : [n] \to [k+1]$ we define $\alpha' : [n+1] \to [k+1]$ to be the map such that $\alpha'|_{[n]} = \alpha$ and $\alpha'(n+1) = k+1$. Note that if the image of α is $[k]$ or $[k+1]$, then the image of α' is $[k+1]$. Consider the family of maps $h_n : E_n \to E_{n+1}$ which maps the summand corresponding to α to the summand corresponding to α' via the identity on A. Let us compute $d_{n+1} \circ h_n - h_{n-1} \circ d_n$. We will first do this in case the category $\mathcal{A}$ is the category of abelian groups. Let us use the notation x_α to indicate the element $x \in A$ in the summand of E_n corresponding to the map α occurring in the index set. Let us also adopt the convention that x_α designates the zero element of E_n whenever $\mathrm{Im}(\alpha)$ is not $[k]$ or $[k+1]$. With these conventions we see that

$$d_{n+1}(h_n(x_\alpha)) = \sum\nolimits_{i=0}^{n+1} (-1)^i x_{\alpha' \circ \delta_i^{n+1}}$$

and

$$h_{n-1}(d_n(x_\alpha)) = \sum\nolimits_{i=0}^{n} (-1)^i x_{(\alpha \circ \delta_i^n)'}$$

It is easy to see that $\alpha' \circ \delta_i^{n+1} = (\alpha \circ \delta_i^n)'$ for $i = 0, \ldots, n$. It is also easy to see that $\alpha' \circ \delta_{n+1}^{n+1} = \alpha$. Thus we see that

$$(d_{n+1} \circ h_n - h_{n-1} \circ d_n)(x_\alpha) = (-1)^{n+1} x_\alpha$$

These identities continue to hold if $\mathcal{A}$ is any abelian category because they hold in the simplicial abelian group $[n] \mapsto \mathrm{Hom}(A, E_n)$; details left to the reader. We conclude that the identity map on E is homotopic to zero, with homotopy given by the system of maps $h'_n = (-1)^{n+1} h_n : E_n \to E_{n+1}$. Hence we see that E is acyclic, for example by Homology, Lemma 0110. □

Lemma 0.3. *Let $\mathcal{A}$ be an abelian category.*

1. *The categories Simp($\mathcal{A}$) and CoSimp($\mathcal{A}$) are abelian.*

(2) *A morphism of (co)simplicial objects $f : A \to B$ is injective if and only if each $f_n : A_n \to B_n$ is injective.*
(3) *A morphism of (co)simplicial objects $f : A \to B$ is surjective if and only if each $f_n : A_n \to B_n$ is surjective.*
(4) *A sequence of (co)simplicial objects*

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is exact at B if and only if each sequence

$$A_i \xrightarrow{f_i} B_i \xrightarrow{g_i} C_i$$

is exact at B_i.

Proof. Pre-additivity is easy. A final object is given by $U_n = 0$ in all degrees. Existence of direct products we saw in Lemmas 0.4 and 0.5. Kernels and cokernels are obtained by taking termwise kernels and cokernels. □

Lemma 0.4. *If U and V are simplicial objects in the category $\mathcal{C}$, and if $U \times V$ exists, then we have*

$$\mathrm{Mor}(W, U \times V) = \mathrm{Mor}(W, U) \times \mathrm{Mor}(W, V)$$

for any third simplicial object W of $\mathcal{C}$.

Proof. Omitted. □

Lemma 0.5. *If U and V are cosimplicial objects in the category $\mathcal{C}$, and if $U \times V$ exists, then we have*

$$\mathrm{Mor}(W, U \times V) = \mathrm{Mor}(W, U) \times \mathrm{Mor}(W, V)$$

for any third cosimplicial object W of $\mathcal{C}$.

Proof. Omitted. □

Lemma 0.6. *Let $\mathcal{A}$ be an abelian category. Let A be an object of $\mathcal{A}$ and let k be an integer. We have $H_i(s(K(A, k))) = A$ if $i = k$ and 0 else.*

Proof. First, let us prove this if $k = 0$. In this case we have $K(A, 0)_n = A$ for all n. Furthermore, all the maps in this simplicial abelian group are id_A, in other words $K(A, 0)$ is the constant simplicial object with value A. The boundary maps $d_n = \sum_{i=0}^{n} (-1)^i \mathrm{id}_A = 0$ if n odd and $= \mathrm{id}_A$ if n is even. Thus $s(K(A, 0))$ looks like this

$$\ldots \to A \xrightarrow{0} A \xrightarrow{1} A \xrightarrow{0} A \to 0$$

and the result is clear.

Next, we prove the result for all k by induction. Given the result for k consider the short exact sequence

$$0 \to K(A, k) \to E \to K(A, k + 1) \to 0$$

from Lemma 0.1. By Lemma 0.3 the associated sequence of chain complexes is exact. By Lemma 0.2 we see that $s(E)$ is acyclic. Hence the result for $k + 1$ follows from the long exact sequence of homology, see Homology, Lemma 0111. □